

GANGSHAN, Taiwan

WMO: 467450

Lat: 22.783N

Lon: 120.262E

Elev: 10

StdP: 101.20

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	11.2	6.0	5.8	11.2	7.7	6.5	12.7	10.2	17.8	9.1	18.6	3.2	0	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	33.2	27.2	32.9	27.1	32.1	26.9	28.2	31.4	27.9	31.2	27.5	30.8	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.2	27.0	22.7	30.1	26.7	22.3	29.8	91.1	31.9	89.6	31.3	87.7	31.0	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.3	6.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			6.1	29.8	1.7	1.0	4.9	30.6	3.9	31.2	2.9	31.7	1.7	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	18.4	19.1	21.5	24.5	26.9	28.3	28.8	28.5	27.7	25.6	22.7	19.4	(6)
(7)		DBStd	4.28	2.33	2.73	2.99	2.52	1.53	1.38	1.14	1.10	1.22	1.61	2.25	2.36	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	79	28	22	9	2	0	0	0	0	0	0	2	16	(9)
(10)		CDD10.0	5224	260	254	356	435	523	548	583	573	532	484	382	292	(10)
(11)		CDD18.3	2262	30	43	107	187	265	298	325	315	282	226	134	49	(11)
(12)		CDH23.3	21842	112	203	648	1438	2604	3460	3941	3696	2993	1841	704	203	(12)
(13)	CDH26.7	7549	6	20	126	406	879	1350	1644	1439	1039	512	111	16	(13)	
(14)	Wind	WSAvg	2.9	3.7	3.6	3.2	2.6	2.5	2.6	2.8	2.7	2.4	2.5	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	1773	20	23	25	76	163	373	340	428	226	53	28	17	(15)
(16)		PrecMax	2837	163	86	104	254	586	1074	838	1168	1079	241	204	137	(16)
(17)		PrecMin	1040	0	0	0	0	1	14	2	93	15	0	0	0	(17)
(18)		PrecStd	496	31	25	30	71	120	268	201	292	228	69	41	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	28.7	30.1	32.2	33.1	34.1	34.2	33.9	33.2	31.3	30.0	28.2	(19)
MCWB			22.4	23.3	24.6	25.3	26.4	26.9	27.4	27.6	27.1	25.8	24.5	23.1	(20)	
2%		DB	25.9	27.1	29.1	31.0	32.1	33.1	33.2	32.9	32.2	30.8	28.9	26.9	(21)	
		MCWB	20.6	22.1	23.9	25.1	26.4	27.0	27.3	27.4	26.8	25.7	24.2	21.8	(22)	
5%		DB	24.6	25.9	28.1	30.0	31.2	32.2	32.8	32.1	31.7	30.0	28.0	25.5	(23)	
		MCWB	19.6	20.9	23.2	24.9	26.2	26.9	27.2	27.1	26.7	25.3	23.4	20.6	(24)	
10%		DB	23.2	24.2	27.0	29.1	30.7	31.8	32.1	31.8	31.0	29.2	27.0	24.2	(25)	
		MCWB	18.5	20.0	22.5	24.5	26.0	26.8	27.0	27.1	26.5	24.7	22.8	19.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	23.8	25.4	26.6	27.9	28.6	28.7	28.6	28.2	26.7	25.6	23.5	(27)
MCD B			26.5	27.4	29.2	30.3	30.9	31.7	31.9	31.9	31.2	30.1	29.4	27.5	(28)	
2%		WB	21.1	22.7	24.4	25.9	27.1	27.9	28.2	28.1	27.6	26.2	24.6	22.2	(29)	
		MCD B	24.7	26.3	28.1	29.6	30.7	31.0	31.6	31.4	30.8	29.7	28.1	25.9	(30)	
5%		WB	20.1	21.5	23.7	25.3	26.6	27.5	27.8	27.7	27.1	25.7	23.9	21.2	(31)	
		MCD B	23.6	25.0	27.3	29.0	30.2	30.7	31.3	31.0	30.4	29.1	27.3	24.6	(32)	
10%		WB	19.2	20.4	23.0	24.7	26.2	27.1	27.4	27.3	26.7	25.2	23.1	20.3	(33)	
		MCD B	22.7	24.0	26.4	28.3	29.5	30.4	30.9	30.5	29.9	28.6	26.2	23.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	9.5	9.1	7.9	7.1	6.3	6.4	6.1	6.5	7.6	8.4	9.8	(35)
MCDBR			10.5	10.6	9.8	8.8	7.8	7.1	7.4	7.0	7.1	7.7	8.5	10.3	(36)	
MCWBR			6.3	6.3	5.8	4.8	4.1	3.5	3.7	3.6	3.7	4.2	4.7	6.0	(37)	
5% WB		MCDBR	9.7	9.9	9.2	8.0	7.3	6.5	6.7	6.5	6.6	7.2	7.9	9.5	(38)	
		MCWBR	6.2	6.3	5.8	4.7	4.2	3.6	3.7	3.6	3.7	4.2	4.9	5.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.507	0.548	0.663	0.650	0.536	0.494	0.497	0.552	0.591	0.601	0.522	0.526	(40)
taud		2.013	1.931	1.695	1.768	2.101	2.248	2.233	2.050	1.912	1.841	2.037	1.974	(41)		
Ebn at Noon		747	745	680	701	780	807	805	764	724	690	727	708	(42)		
Edh at Noon		159	184	243	229	163	140	142	170	192	198	155	161	(43)		
(44)	All-Sky Solar Radiation	RadAvg		3.39	3.94	4.33	4.78	5.11	5.36	5.71	5.09	4.96	4.55	3.65	3.24	(44)
(45)		RadStd		0.36	0.44	0.40	0.49	0.57	0.58	0.47	0.56	0.43	0.28	0.29	0.33	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.24	N/A	N/A	N/A	+0.20	+0.24	N/A	N/A	+116	+110

Nomenclature: See separate page